

Table 1: The 30 viruses exhibiting the highest number of documented interactions with human hosts.

Virus ID	Freq	Virus ID	Frequency	Virus ID	Freq
Q77M19	438	Q1K9H5	238	P09992	184
P04608	371	B9UPF3	233	A0A024B7W1-PRO_0000443029	183
P06428	327	P04618	228	P04591	181
P04578	305	P0DOE9	219	P06461	180
P06460	278	P0CK57	203	P03070	166
P29990-PRO_0000037929	273	P03496	202	P06463	164
P03129	268	P04620	198	P0C6Z1	164
P06792	261	G3CKS7	195	P06927	160
P0C737	251	Q2MG95	195	Q32ZE1-PRO_0000435839	160
Q20MH8	239	Q9WMX2-PRO_0000037548	195	P29991-PRO_0000037936	154

Table 2: Number of features extracted using the three aforementioned encoding techniques

Feature Extraction Method	Virus Features	Human Features	Total
Conjoint Triad	343	343	686
Amino Acids Sequences	20×protein length	20×protein length	various length
Gene Ontology (GO) terms	609	18405	19014

Table 3: The results derived from applying various dimensionality reduction techniques to the CT features of the NearMiss-balanced dataset

	CT (NearMiss)	PCA	MCFS	NMF	FA	Optuna	ISUD
Accuracy	92.65	88.91	91.23	91.09	87.93	91.02	92.91
Precision	92.25	90.40	92.22	92.11	89.96	91.95	92.51
Recall	93.14	87.06	90.06	89.87	85.39	89.90	93.40
F1-Score	92.69	90.76	91.13	90.98	87.62	90.92	92.95
Specificity	92.15	88.70	92.40	92.30	90.47	92.13	92.41
MCC	85.30	77.87	82.48	82.20	75.96	82.06	85.83
ROC AUC	95.36	95.28	96.98	96.99	94.62	96.88	96.16
PR AUC	96.55	95.73	97.19	97.17	95.05	97.09	94.29

Table 4: Computational characteristics and predictive performance of the three teacher models and the knowledge-distilled (KD) student model.

Measure	Attention Teacher	FPN Teacher	GLU Teacher	KD Student
Parameters	27,394	310,285	589,962	2,817,282
Trainable Parameters	0	0	0	2,817,282
Model Size (MB)	0.1045	1.1836	2.2505	10.7488
Training Time (s)	414.1246	297.1828	262.6379	776.6857
Inference Time (ms/sample)	0.027765	0.021404	0.018641	0.036547
Peak Memory (MB)	17.8726	23.4805	31.3438	98.3467
Accuracy (%)	89.81	90.80	89.74	92.91
Precision (%)	91.13	91.55	92.31	92.51
Recall (%)	88.21	89.89	86.69	93.40
F1 (%)	89.65	90.71	89.41	92.95
Specificity (%)	91.41	91.70	92.78	92.41
MCC (%)	79.67	81.61	79.62	85.83
AUC (%)	91.89	94.45	93.78	96.16

Table 5: Comparative analysis between the proposed model and selected state-of-the-art methods.

Authors	Feature Extraction	Feature Selection	Balancing Technique	Prediction Method	ACC	F1	MCC	AUC
Ibrahim et al. (2023)	CT	SVM	-	RF	83.40	40.60	41.20	90.70
Cihan et al. (2025)	GO	MMPC	-	NLP-DNN	-	79.30	-	87.80
Yang et al. (2020)	Doc2Vec	-	-	RF	79.17	79.79	58.40	87.10
Our model	CT, GO, AA Seq	ISUD	NearMiss	Hybrid DL	92.91	92.95	85.83	96.16

Supplementary Table 1: The Conjoint Triad results on 25 datasets using our proposed model.

Dataset	Accuracy	Precision	Recall	F1-Score	Specificity	MCC	PR	ROC
Data_1	86.40	86.08	86.80	86.44	86.00	72.81	92.32	93.28
Data_2	85.92	84.03	88.72	86.31	83.18	72.01	91.68	92.79
Data_3	85.03	85.17	84.79	84.98	85.27	70.07	90.69	92.09
Data_4	85.41	84.68	86.40	85.53	84.41	70.83	91.03	92.27
Data_5	86.39	84.93	88.44	86.65	84.35	72.85	91.28	92.48
Data_6	86.37	86.34	86.36	86.35	86.37	72.74	91.75	93.s02
Data_7	86.53	86.01	87.22	86.61	85.84	73.08	91.46	92.89
Data_8	86.43	86.29	86.58	86.44	86.29	72.87	92.21	93.16
Data_9	86.08	85.94	86.23	86.08	85.93	72.16	90.83	92.62
Data_10	86.66	86.07	87.44	86.75	85.89	73.34	92.30	93.04
Data_11	85.98	84.35	88.30	86.28	83.66	72.05	91.55	92.57
Data_12	85.89	86.34	85.23	85.78	86.55	71.79	91.51	92.44
Data_13	85.38	84.48	86.65	85.55	84.13	70.80	90.94	92.34
Data_14	86.61	86.49	86.74	86.61	86.48	73.22	91.80	92.97
Data_15	86.43	85.96	87.04	86.50	85.82	72.88	92.02	93.01
Data_16	85.73	84.72	87.13	85.91	84.32	71.49	91.39	92.53
Data_17	86.02	85.83	86.25	86.04	85.80	72.06	91.17	92.43
Data_18	85.99	86.53	85.21	85.87	86.77	72.00	91.23	92.48
Data_19	86.34	86.54	86.03	86.29	86.66	72.70	92.07	92.99
Data_20	86.42	86.52	86.25	86.38	86.59	72.85	90.92	92.52
Data_21	86.89	86.10	87.93	87.01	85.84	73.79	91.83	93.01
Data_22	87.14	86.80	87.55	87.18	86.73	74.29	92.07	93.33
Data_23	86.60	85.47	88.15	86.79	88.05	73.24	92.26	93.06
Data_24	86.28	87.14	85.08	86.10	87.48	72.58	91.43	92.71
Data_25	86.11	87.15	94.30	90.58	66.15	64.97	93.67	89.78

Supplementary Table 2: The amino acid sequence-based features result on 25 datasets using our proposed model.

Dataset	Accuracy	Precision	Recall	F1-Score	Specificity	MCC	PR	ROC
Data_1	50.51	50.42	52.15	51.27	48.86	40.38	63.23	50.50
Data_2	50.40	50.33	49.08	49.70	51.70	39.53	62.42	50.39
Data_3	50.21	50.14	50.43	50.28	49.98	41.99	62.66	50.20
Data_4	49.66	49.59	50.05	49.82	49.26	48.31	62.29	49.65
Data_5	49.61	49.53	47.77	48.63	51.44	40.25	61.69	49.61
Data_6	49.46	49.38	48.77	49.07	50.14	42.48	61.86	49.45
Data_7	50.04	49.97	50.25	50.11	49.83	43.29	62.53	50.04
Data_8	51.16	51.26	44.46	47.62	57.83	43.82	61.72	51.14
Data_9	51.07	50.94	54.18	52.51	47.96	41.01	64.00	51.07
Data_10	49.49	49.39	46.69	48.00	52.28	44.45	61.35	49.48
Data_11	49.27	49.22	50.29	49.75	48.24	40.90	62.16	49.27
Data_12	50.53	50.46	50.49	50.48	50.56	41.78	62.83	50.52
Data_13	50.52	50.43	52.77	51.57	48.26	40.65	63.39	50.52
Data_14	49.59	49.50	47.64	48.55	51.53	41.46	61.64	49.58
Data_15	49.37	49.31	50.20	49.75	48.53	42.36	62.19	49.37
Data_16	49.81	49.74	50.43	50.08	49.19	44.18	62.46	49.81
Data_17	50.36	50.28	51.84	51.05	48.88	43.57	63.09	50.36
Data_18	50.21	50.15	47.46	48.77	52.94	41.27	61.92	50.20
Data_19	50.74	50.63	53.54	52.05	47.93	48.84	63.69	50.74
Data_20	51.53	51.46	51.62	51.54	51.44	45.24	63.62	51.53
Data_21	50.58	50.51	51.31	50.90	49.85	44.38	63.06	50.58
Data_22	50.61	50.54	50.58	50.56	50.62	46.76	62.90	50.60
Data_23	50.01	49.94	48.75	49.34	51.26	39.82	62.14	50.00
Data_24	49.94	49.87	49.45	49.66	50.42	40.14	62.28	49.94
Data_25	50.08	50.21	50.67	49.92	50.20	44.69	61.34	49.98

Supplementary Table 3: The GO terms result on 24 datasets using our proposed model.

Dataset	Accuracy	Precision	Recall	F1-Score	Specificity	MCC	PR	ROC
Data_1	67.53	68.39	64.43	66.35	70.60	35.10	72.58	73.59
Data_2	81.33	80.37	82.59	81.47	80.09	62.70	86.62	87.76
Data_3	80.92	79.90	82.29	81.08	79.57	61.87	85.22	87.54
Data_4	80.49	79.29	82.44	80.83	78.54	61.03	85.07	87.11
Data_5	81.16	81.87	79.73	80.78	82.56	62.33	86.97	87.87
Data_6	81.78	82.87	79.83	81.32	83.71	63.61	86.18	87.98
Data_7	80.96	80.06	82.11	81.07	79.82	61.94	86.12	87.56
Data_8	81.45	80.45	82.77	81.59	80.14	62.93	86.79	88.44
Data_9	80.65	79.12	82.94	80.99	78.39	61.39	85.51	87.16
Data_10	80.70	81.53	79.07	80.28	82.31	61.43	86.19	87.39
Data_11	80.29	84.31	74.12	78.89	86.38	60.99	86.94	87.49
Data_12	80.59	79.77	81.63	80.69	79.57	61.21	85.41	87.31
Data_13	80.80	82.96	77.23	79.99	84.34	61.74	85.60	87.34
Data_14	81.36	80.96	81.68	81.32	81.04	62.72	86.89	87.86
Data_15	81.26	82.30	79.33	80.79	83.16	62.55	85.33	87.49
Data_16	80.48	79.76	81.35	80.55	79.62	60.98	84.49	86.56
Data_17	82.27	81.77	82.77	82.27	81.79	64.56	87.40	88.68
Data_18	80.98	83.29	77.20	80.13	84.71	62.11	87.19	88.15
Data_19	81.65	81.66	81.33	81.49	81.96	63.30	87.07	88.00
Data_20	83.03	83.62	81.88	82.74	84.16	66.07	87.34	89.41
Data_21	80.68	78.17	84.79	81.34	76.62	61.60	85.87	87.84
Data_22	81.58	84.82	76.65	80.53	86.46	63.44	87.10	88.32
Data_23	81.71	83.57	78.64	81.03	84.74	63.52	87.03	88.24
Data_24	82.28	83.02	83.47	83.43	80.51	64.40	87.44	88.51

Supplementary Table 4: Results obtained from ten runs of the ISUD feature selection algorithm.

Run	Seed	Best_Fitness	Best_Validation_Accuracy_Percent	Selected_Features
1	42	91.99	91.99	429
2	123	92.91	92.91	340
3	456	91.96	91.96	425
4	789	91.88	91.88	327
5	1024	92.67	92.67	406
6	2048	91.95	91.95	378
7	3072	91.94	91.94	347
8	4096	91.95	91.95	426
9	5120	92.14	92.14	437
10	6144	91.87	91.87	408